

devices," says Thakurdesai. This will help embedded engineers to select a reliable device at an affordable price for their application.

Re-programmability is one of the biggest benefits of flash-based MCUs. But, Eieland points out, "We are approaching a flash challenge for smaller flash densities, as we get closer and closer to 45 and 90 nm for MCUs with less than 64 kB of flash. There are also potential issues with 5V support at a reasonable cost with these processes." The industry is working towards newer technologies to overcome this challenge.

As the technology advances, the processes become more and more complex. Justifying this with an example, Duvenhage says, "Smaller geometry manufacturing processes imply lower cost for digital and memory modules, but higher complexity and cost for achieving good analogue and low-power performance. This implies that the overall cost reduction of the product depends very much on the type of product that is being developed or the potential markets that can be addressed with these products." The end result is more complex process technologies, more expensive development cycles and more specialised products that ultimately require larger markets to justify the investment required for their development.

MCUs are also facing threats from the new entrants in programmable logic devices like CPLDs and FPGAs. Thapliyal says, "Although programmable logic devices have their own limitations to take over larger MCU market share, these devices possess better input/output management than MCUs and thus can have shorter time to adapt to changing requirements of the projects."

In a market with traditionally available choices in 4-bit, 8-bit and 16-bit designs, there is a growing demand to come up with efficient solutions that retain the ability for greater analogue integration and faster time-to-market. Thakurdesai feels that this need is also

MCU Manufacturing in India

"India is rich in talented software engineers and has good R&D facilities, but it still has a long way to go before it becomes a semiconductor manufacturing hub," states an article titled "Can India be a viable place for manufacturing semiconductors?" published in Electronics Bazaar magazine. India has emerged as a destination for product manufacturing with several original equipment manufacturers (OEMs) setting up their manufacturing units in India. However, semiconductor manufacturing is quite capital-intensive and requires huge infrastructural support, states the analysis.

India has great potential to grow its semiconductor manufacturing capabilities, but this requires investments in excess of \$3-5 billion and a well-developed ecosystem and education. Fabs also need co-location of all the raw material suppliers in the same area near the manufacturing plant to be more efficient. Access to ports, uninterrupted electricity, abundant supply of clean water, waste management capability, warehousing, good road network and global air connectivity are critical for success. Thus it is advisable to boost the electronics design and manufacturing climate in India via a vis fabs, promote product development, nurture startups and introduce a level-playing field at different points in the entire electronics ecosystem.

being driven by emerging areas like LED lighting, smart metering, energy harvesting, consumer electronics and cutting-edge medical applications that emphasise on lower cost, smaller size and higher performance. He adds, "The ultra-low-cost microcontroller market has traditionally been the domain of low-end 8-bit MCUs, while 16-bit MCUs having better performance have been seen as too expensive. In recent times, there has been a need for increased performance in the growing low-end 8-bit market."

"Another major constraint and challenge today that we see is cost, which can be addressed only if India's semiconductor manufacturing ecosystem gets developed," feels Ivaturi.

MCU next

Embedded systems are in need of more and more MCUs for a variety of existing and emerging applications. As indicated earlier, the MCUs are becoming more and more complex in terms of specific functionalities and performances. "Semiconductor firms are continuously improving their product portfolio by increasing the performance and addressing the segment-targeted needs," shares Thapliyal.

Touch solutions is another high-research area. "Touch phones and tablets are a market with a growth rate we have never seen before. MCU vendors are working towards providing

more efficient and innovative solutions for touch-based applications" shares Eieland.

Peripheral enhancement and low-power features are the main focus for many companies to enrich the general-purpose MCU portfolio. Thapliyal says, "Especially for applications like home/building automation, smart metering, healthcare and consumer electronics, MCUs are featured with interfaces like RF ZigBee." You may expect a lot of other integrated features for different RF standards to enhance the usability of MCUs and reduce time-to-market. Computing performance enhancement is another area that the companies are working on to further reduce the gap with digital signal processors (DSPs).

Many emerging applications, such as medical electronics and handheld electronic gadgets, are being designed with communication capabilities to provide continuous data flow for recording and monitoring. Such applications demand advanced technologies that allow greater miniaturisation to further improve portability and functionality. "Portability and miniaturisation will be a major step ahead in creating innovative solutions and helping medical electronics become more flexible, affordable and accessible," feels Thakurdesai. ●

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